

# Ivar Cashin Eriksson

## Curriculum Vitae

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### Introduction

Data scientist and ML engineer with seven years of applied experience and an academic background in Engineering Physics and Statistical Learning. My work has focused on numerical modelling, optimisation, probabilistic methods, and machine learning. I am interested in applying this background to challenging problems in quantum computing.

### Education

- 2015–2020 **Royal Institute of Technology, KTH – Data Science and Engineering**, GPA: 5.0 (out of 5.0)  
Master: Statistical Learning and Data Analytics, Bachelor: Engineering Physics.  
Relevant coursework: Quantum Physics; Solid State Physics; Electromagnetic Theory.  
Thesis: [Probabilistic Dose Prediction for Radiation Therapy Planning](#)
- 2019–2019 **Swiss Federal Institute of Technology, ETH – Applied Mathematics**, *Exchange*
- 2017–2022 **Stockholm University, SU – Economics**, *Supplementary Bachelor's Degree*
- 2017–2017 **Zhejiang University (ZJU) – Exchange**, *Optics laboratory work*, Hangzhou, China  
Work in optics laboratory on laser trimming microring resonators, in collaboration with a local research group. Fabricated microring resonators and tuned their resonance frequency by applying controlled laser pulses to selectively remove cladding material for integrated photonic circuits.

### Selected Work Experience

Researcher within Data Science @ RaySearch Laboratories

- Developed ML methods for automated radiotherapy planning from segmented 3D CT scans.
- Applied autoencoders, Gaussian processes, probabilistic modelling, and multi-objective optimisation.
- Translated clinical constraints into mathematical models in collaboration with medical physicists.

*Keywords: 3D computer vision, probabilistic modelling, latent representation, segmentation, autoencoders, optimisation, healthcare AI*

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Full-Stack Data Scientist & Project Manager @ Signum Life Science

- Leading development of a production forecasting platform for pharmaceutical prices and demand.
- Working across mathematical modelling, software architecture, data engineering, and ML engineering.

*Keywords: Explainable AI, Azure Databricks, forecasting, XGBoost, pricing, ML engineering, project management*

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Principal Data Scientist @ Valcon

- Architected an end-to-end training and inference system for a real-time neural network.
- Led technical delivery across modelling, software architecture, and production deployment.
- Mentored data scientists in reproducibility, software design, explainable AI, and Git.

*Keywords: Real-time ML, NLP, explainability in AI, MLOps, multi-modal, Azure, Docker, mentoring, stakeholder management*

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### Skills and Other

- Technology Python, SQL, C++, PyTorch, TensorFlow, MLflow, Databricks, MS Azure
- Modelling Optimisation, probabilistic modelling, sparse Gaussian processes, deep learning.
- Language *Native:* English, Swedish. *Intermediate:* Danish.
- Other Valcon People's Choice Award (2023); IYPT Sweden (2015); Mensa Sweden.

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## Detailed Work Experience

### 2024–Present **Full-Stack Data Scientist and Project Manager**, *Signum Life Science*, Copenhagen

At Signum, my current work is on an end-to-end Databricks forecasting platform for pharmaceutical price and demand forecasting. I work across architecture, project management, data science, ML engineering, and software engineering for the platform. The work builds on a pharmaceutical price forecasting model that helps clients navigate Denmark's fortnightly blind auctions.

Earlier in the role, I owned and extended Kwarts, a next-best-action engine for pharmaceutical sales representatives. Originally built externally, the system was extended with features such as explainable AI in collaboration with UX designers, and supported commercialisation by aiding our sales representatives. Kwarts adapts to varied customer data maturity, from rule-based logic to deep learning, and is deployed in AWS for scalability.

Beyond model development, I organised an internal week-long hackathon that included up to 40 participants.

### 2022–2024 **Principal Data Scientist**, *Valcon*, Copenhagen

Led data science projects across pharma, energy, IoT, and manufacturing. Mentored junior colleagues in software architecture and data science best practices, led explainable AI and Git trainings, and regularly presented complex technical topics to non-technical stakeholders.

One project involved training a real-time deep neural network with a custom training pipeline and modular architecture. The training data consisted of natural-language inputs and high-dimensional multi-class labels, which made for a challenging prediction problem. I was the lead data scientist and architect of the solution and responsible for implementing the end-to-end production pipeline.

I was awarded the "Valcon People's Choice Award" for exemplary work, leadership, team spirit, and scientific curiosity.

### 2021–2022 **Data Scientist**, *Valtech*, Copenhagen

Delivered ML solutions in e-commerce. Built time series forecasts using Facebook Prophet, and applied clustering to behavioural data to support personalisation. Worked closely with business stakeholders to ensure actionable outputs.

### 2019–2021 **Researcher within Data Science**, *RaySearch Laboratories*, Stockholm

At RaySearch Laboratories, I worked on automating radiotherapy planning using a combination of deep learning, probabilistic modelling, and multi-objective optimisation. I developed a pipeline that extracted latent anatomical features from segmented 3D CT scans using autoencoders, and used these representations to identify similar patients via a learned similarity metric. Dose-volume histograms and spatial dose distributions were then predicted using sparse Gaussian processes, providing a probabilistic framework for dose planning conditioned on patient geometry.

Alongside this, I redesigned some of RaySearch's lexicographic optimisation algorithms to better reflect clinical decision hierarchies, particularly in palliative care where trade-offs between tumour control and organ sparing are nuanced. I collaborated closely with medical physicists and clinicians throughout, ensuring the models aligned with oncological standards and practical workflow requirements.

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## Other Detailed Experience

### 2025–2025 **Solo Project**, *Project Iris*

In this personal project I built a proof-of-concept end-to-end computer vision system for making product images on e-commerce sites interactive. The pipeline detects individual items using a YOLOv8 model, and embeds them semantically using OpenCLIP. The embeddings are then compared to support intelligent product linking across all images and products in a web store.

The system includes a full-stack setup with a FastAPI backend, MongoDB for persistence, a Qdrant index for vector search, and a custom-built Chromium plug-in that overlays clickable links directly onto web shop imagery in real time. The project also includes a dynamic scraping and indexing pipeline for ingesting and structuring product data from arbitrary e-commerce sites.

The project was a learning opportunity for fine-tuning of deep learning models and practical ML engineering.